ModelIt!

Student Activity Pack

How Do We Know the Earth Is 4.5 Billion Years Old?

Modeling Radioactive Decay, Radiometric Dating, and Deep Time

Grade 10L1 | NGSS HS-ESS1-5, HS-ESS1-6 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think scientists figured out the age of the Earth if no one was around to see it form?

2 What do you know about radioactivity and how it might be used as a clock?

3 Draw a diagram showing what you think happens to radioactive atoms over very long periods of time.

4 Why do you think scientists say the Earth is 4.5 billion years old and not 4 billion or 5 billion?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Radioactive Isotope Amount | Decay Rate | Rock Layer Age | Geological Event Timing | Age Estimate Accuracy

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Young Rock

Parent:Daughter ratio high | Short half-life isotope

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Ancient Rock

Parent:Daughter ratio low | Long half-life isotope

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Multiple Methods

Same rock | Three different isotope systems

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Isochron Analysis — A graphical method that plots multiple minerals from the same rock to determine age without knowing the initial parent-daughter ratio — provides a built-in check on whether the system remained closed

Concordance Testing — Using two independent decay chains (U-235 to Pb-207 and U-238 to Pb-206) from the same sample — if both give the same age, the result is 'concordant' and highly reliable

Metamorphic Resetting — The process where heat and pressure can partially or completely reset a rock's radiometric clock by allowing isotopes to escape or re-equilibrate — dating metamorphic rocks requires understanding these resetting events

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can scientists confidently claim the Earth is 4.5 billion years old when no human has ever seen anything close to that old? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that radioactive decay is a reliable clock?**

2. **Why is the convergence of multiple independent dating methods such powerful evidence for Earth's age?**

3. **What would it mean if different dating methods gave significantly different ages for the same rock?**

4. **How does understanding radiometric dating change your perception of time and Earth's history?**

5. **What are the limitations of modeling radioactive decay with only five components?**

8 STEM Challenge: Design a Rock Dating Laboratory Protocol

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YOUR ENGINEERING MISSION

Design a laboratory protocol for dating an unknown rock sample using multiple radiometric methods, including quality control measures to ensure accuracy.

THE SCENARIO

A geological survey team has discovered rock formations that may contain evidence of an ancient mass extinction event. They need to determine the precise age of these rocks to correlate them with known extinction events. Your team must design the dating protocol — which isotope systems to use, how to prepare samples, what quality control checks to perform, and how to interpret the results.

Engineering Design Thinking

1. WHAT ISOTOPE SYSTEMS WILL YOU ANALYZE AND WHAT IS THE SCIENTIFIC BASIS FOR EACH CHOICE?
2. HOW WILL YOU PREPARE ROCK SAMPLES TO AVOID CONTAMINATION OF PARENT OR DAUGHTER ISOTOPES?
3. WHAT STATISTICAL METHODS WILL YOU USE TO CALCULATE AGE AND UNCERTAINTY FROM YOUR MEASUREMENTS?
4. HOW WILL YOU DETERMINE WHETHER THE ROCK SYSTEM HAS REMAINED 'CLOSED' SINCE FORMATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Geochronologists specialize in determining the age of rocks, minerals, and geological events using radiometric and other dating techniques. They work at universities, geological surveys, and mining companies, earning $65,000–$120,000/year. Isotope Geochemists who develop new dating methods and study planetary formation earn $70,000–$140,000/year.